

## Miscellaneous Theorems and Ideas

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*This is just a collection of some cool facts and theorems I have found, along with their proofs.*

### Theorem 1.

For an odd positive integer  $n$ , and positive integers  $a, b$  with  $a^2 \leq b < (a+1)^2$  we have the following,

$$\left\lfloor (a + \sqrt{b})^n \right\rfloor = (a + \sqrt{b})^n + (a - \sqrt{b})^n.$$

### Proof of Theorem 1.

Firstly, due to Binomial expansion,

$$\begin{aligned} (a + \sqrt{b})^n + (a - \sqrt{b})^n &= \sum_{r=0}^n \binom{n}{r} (\sqrt{b})^r a^{n-r} + \sum_{r=0}^n \binom{n}{r} (-\sqrt{b})^r a^{n-r} \\ &= \sum_{r=0}^n \binom{n}{r} (\sqrt{b})^r [1 + (-1)^r] a^{n-r} \\ &= 2 \sum_{\substack{r=0 \\ r \text{ even}}}^n \binom{n}{r} (\sqrt{b})^r a^{n-r} \end{aligned}$$

which is clearly an integer. Now it suffices to show that

$$(a + \sqrt{b})^n - 1 < (a + \sqrt{b})^n + (a - \sqrt{b})^n \leq (a + \sqrt{b})^n$$

That is

$$-1 < (a - \sqrt{b})^n \leq 0$$

but we know that  $a \leq \sqrt{b} < a+1$  and so  $a - \sqrt{b} \leq 0$  and  $-1 < a - \sqrt{b}$  and raising both inequalities to the power of  $n$  gives the desired result. ■

**Theorem 2.**

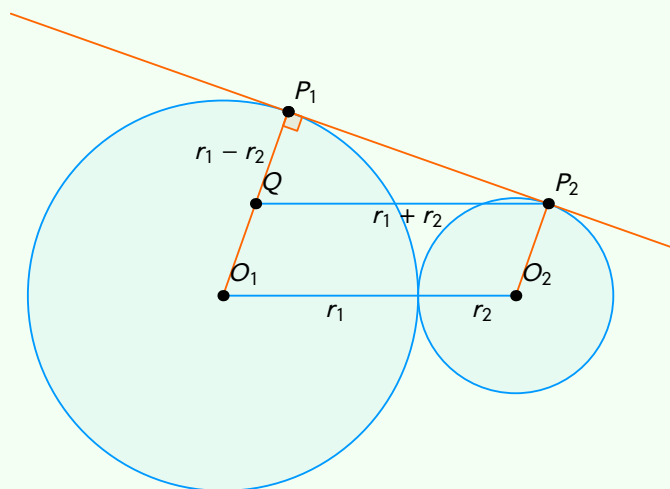
Let  $C_1$  and  $C_2$  be externally tangent circles with centres  $O_1, O_2$  and radii  $r_1, r_2$ , ( $r_1 \geq r_2$ ) respectively. Let  $P_1, P_2$  be points on  $C_1, C_2$  respectively such that  $P_1P_2$  is a common tangent to both circles.

(a) Show that  $P_1P_2 = 2\sqrt{r_1r_2}$ .

(b) Find the area of the region bounded by the two circles and their common tangent.

**Solution**

(a) The idea is to draw in the line  $l$  through  $P_2$  that is parallel to  $O_1O_2$ . Let  $Q = l \cap O_1P_1$ .



We notice that since  $O_1P_1$  and  $O_2P_2$  are radii they meet the common tangent at right angles, and in particular  $O_1P_1 \parallel O_2P_2$ . Now it is clear to see that  $QP_2 = r_1 + r_2$  and  $QP_1 = r_1 - r_2$ . Now owing to Pythagoras,

$$P_1P_2 = \sqrt{(r_1 + r_2)^2 - (r_1 - r_2)^2} = 2\sqrt{r_1r_2}. \quad \square$$

(b) Notice that due to the parallel conditions  $O_1P_1P_2O_2$  is a trapezium, with area

$$\frac{r_1 + r_2}{2} (2\sqrt{r_1r_2}).$$

Now we just need to subtract off the two sectors. Let  $\angle P_1O_1O_2 = \theta$ . Notice that  $\angle O_1O_2P_2 = 180 - \theta$  due to corresponding angles. Taking  $\theta$  to be measured in radians gives the area of the two sectors to be

$$\frac{1}{2}r_1^2\theta + \frac{1}{2}r_2^2(\pi - \theta) = \frac{1}{2}\theta(r_1^2 - r_2^2) + \frac{1}{2}\pi r_2^2.$$

So the final area is

$$(r_1 + r_2)\sqrt{r_1r_2} - \frac{1}{2}\theta(r_1^2 - r_2^2) + \frac{1}{2}\pi r_2^2. \quad \blacksquare$$

*Comments: I first came across the fact (a) from a BPRP (Black pen red pen) video, and then I actually used this fact to solve a problem in a Year 12/13 geometry topic test. Part (b) actually came up as a question with special case  $r_1 = 3, r_2 = 1$  when I was sitting the 2026 Jan TMUA, I was unable to solve it during contest.*

**Theorem 3.**

Consider some  $3 \times 3$  matrix  $A$  of the form

$$A = \begin{bmatrix} k & 0 & 0 \\ x & k & 0 \\ y & z & k \end{bmatrix}.$$

Let  $M$  be another  $3 \times 3$  matrix defined like so,  $A = kI + M$ . Then

$$A^n = k^n I + k^{n-1} nM + k^{n-2} \frac{n(n-1)}{2} M^2.$$

**Solution**

*Claim 1:  $M$  is nilpotent with nilpotency index  $\leq 3$ . In other words there is a positive integer  $k \leq 3$  such that  $M^k = 0$ .*

*Proof:* We use the Caley-Hamilton theorem. Look at the characteristic equation of  $M$ ,

$$\begin{vmatrix} -\lambda & 0 & 0 \\ x & -\lambda & 0 \\ y & z & -\lambda \end{vmatrix} = -\lambda^3 = 0.$$

Now since  $M$  satisfies it's own characteristic equation (by Caley-Hamilton) we have  $M^3 = 0$ .  $\square$

Now we can apply Binomial theorem like so,

$$\begin{aligned} A^n &= (kI + M)^n \\ &= \sum_{r=0}^n \binom{n}{r} (M)^r (kI)^{n-r} \\ &= \binom{n}{0} k^n I + \binom{n}{1} k^{n-1} M + \binom{n}{2} k^{n-2} M^2 \\ &= k^n I + n k^{n-1} M + \frac{n(n-1)}{2} k^{n-2} M^2 \end{aligned}$$

where the expansion stops at  $r = 2$  because any further terms will have a factor of  $M^3 = 0$ .  $\blacksquare$

**Theorem 4.**

Take  $2^n + 1$  lattice points in  $n$ -dimensional space. Then there are two points  $P, Q$  out of this group such that the segment with endpoints  $P, Q$  passes through another point with integer coordinates.

**Solution**

Consider a point from this group

$$P = (p_1, p_2, \dots, p_n).$$

There are  $2^n$  ways to choose each of the parities of each of the  $p_i, (i = 1, 2, \dots, n)$ . Since we have  $2^n + 1$  points, by Pigeonhole principle there must be some point  $Q$  also a part of this group

$$Q = (q_1, q_2, \dots, q_n)$$

that satisfies

$$p_i \equiv q_i \pmod{2}, \quad (i = 1, 2, \dots, n).$$

Now define

$$\vec{\Delta} = \vec{OQ} - \vec{OP} = \langle x_1, x_2, \dots, x_n \rangle$$

where  $x_i = q_i - p_i$ . Note that  $x_i$  is even because  $p_i$  and  $q_i$  have the same parity. Since every component is even

$$\frac{1}{2}\vec{\Delta} \in \mathbb{Z}^n$$

and so the point at

$$\vec{OP} + \frac{1}{2}\vec{\Delta} = \left( p_1 + \frac{1}{2}x_1, p_2 + \frac{1}{2}x_2, \dots \right)$$

is a lattice point and lies on the segment  $PQ$  between  $P$  and  $Q$ . ■

**Theorem 5 — (Strengthened Theorem 1).**

Let  $a, b \in \mathbb{N}$ ,  $b$  not a perfect square. Define  $x = a + \sqrt{b}$  and  $y = a - \sqrt{b}$ . Let  $n \in \mathbb{N}$ .

- If  $0 < y < 1$ , then  $x^n + y^n = \lceil x^n \rceil$ .
- If  $-1 < y < 0$  and  $n$  is odd, then  $x^n + y^n = \lfloor x^n \rfloor$ .

**Solution**

We present a strengthened version of theorem 1, along with motivation for why these statements are true. Define  $S_n = x^n + y^n$ . First we are going to show that  $S_n \in \mathbb{Z}$ . Then since  $|y| < 1$  we know that  $y^n$  is very small, enough to restrict  $S_n$  to an interval of size  $< 1$ . Finally we are able to conclude the results of theorem 5.

Note that  $x, y$  are the roots of

$$x^2 - 2ax + (a^2 - b) = 0.$$

If one has studied how to solve second order linear recurrence relations then you know that it is the case that

$$S_{n+1} = 2aS_n - (a^2 - b)S_{n-1}$$

with the initial conditions  $S_0 = 2, S_1 = 2a$  both of which are integers. Since  $a, b \in \mathbb{N}$  and the integers are closed under  $+, -, \times$  we deduce that  $S_n \in \mathbb{Z}$ .

Now observe that if  $0 < y < 1$  then  $0 < y^n < 1$ . Thus  $S_n$  satisfies the inequality

$$x^n < S_n < x^n + 1.$$

Now since  $S_n$  is an integer in the interval  $(x^n, x^n + 1)$  it is the smallest integer  $> x^n$ , that is  $S_n = \lceil x^n \rceil$ .

Similarly if  $-1 < y < 0$  then  $-1 < y^n < 0$  since  $n$  is odd. So  $S_n$  satisfies

$$x^n - 1 < S_n < x^n.$$

Again since  $S_n$  is an integer in the interval  $(x^n - 1, x^n)$  it is the largest integer  $< x^n$ , that is  $S_n = \lfloor x^n \rfloor$ . ■